

Osptek Display

LCD TFT SPECIFICATION

Model No:

LCD12864-1937

osptek[®]

1. REVISION HISTORY

The logo for osptek, featuring a stylized 'X' shape composed of two overlapping triangles, one light blue and one light purple, set against a light gray grid background. The word 'osptek' is written in a bold, lowercase, sans-serif font, with a registered trademark symbol (®) to its upper right.

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3. FEATURES

1. Support display capacity: 128x 64 Pixels dots with FSTN mode.
2. On-chip Display Data RAM , Capacity: $128 \times 64 = 8192$ bits.
3. Serial interfaces are supported.
4. Abundant command functions Display data Read/Write, display ON/OFF, Normal/Reverse display mode, page address set, display start line set, column address set, status read, display all points ON/OFF, LCD bias set, electronic volume, read/modify/write, segment driver direction selects, power saver, static indicator, common output status select, Vo voltage regulation internal resistor ratio set.
5. Static drive circuit equipped internally for indicators.(1 system, with variable flashing speed.)
6. Low-power liquid crystal display power supply circuit equipped internally. Booster circuit (with Boost ratios of 4X/5X, where the step-up voltage reference power supply can be input externally).
7. High-accuracy voltage adjustment circuit (Thermal gradient $-0.05\%/^{\circ}\text{C}$) V0 voltage regulator resistors equipped internally, LCD voltage divider resistors equipped internally, electronic volume function equipped internally, voltage follower.
8. CR oscillator circuit equipped internally.

4. GENERAL SPECIFICATIONS

ITEM	DESCRIPTION	UNIT
Outline Size	$37.2(\text{L}) \times 39.2(\text{W}) \times 5.2(\text{T})$	mm
LCD Type	FSTN, Transflective / Positive, 1/65Duty, 1/9Bias	---
Display type	128×64 dots	---
View Area	32.0×29.0	mm
Display Area	29.41×27.49	mm
Dots size	0.20×0.40	mm
Dots pitch	0.23×0.43	mm
Controller & driver	ST7567-COG	---
View Direction	6 O'Clock	---
Interface mode	8 bit 6800, 4-Line Interface	---
VDD&VOP(Type)	3.3 V & 8.8 V	V
Backlight(Type)	White, 3.5V, 14mA	---
Operation Temp.	$-20 \sim +70$	$^{\circ}\text{C}$
Storage Temp.	$-30 \sim +80$	$^{\circ}\text{C}$

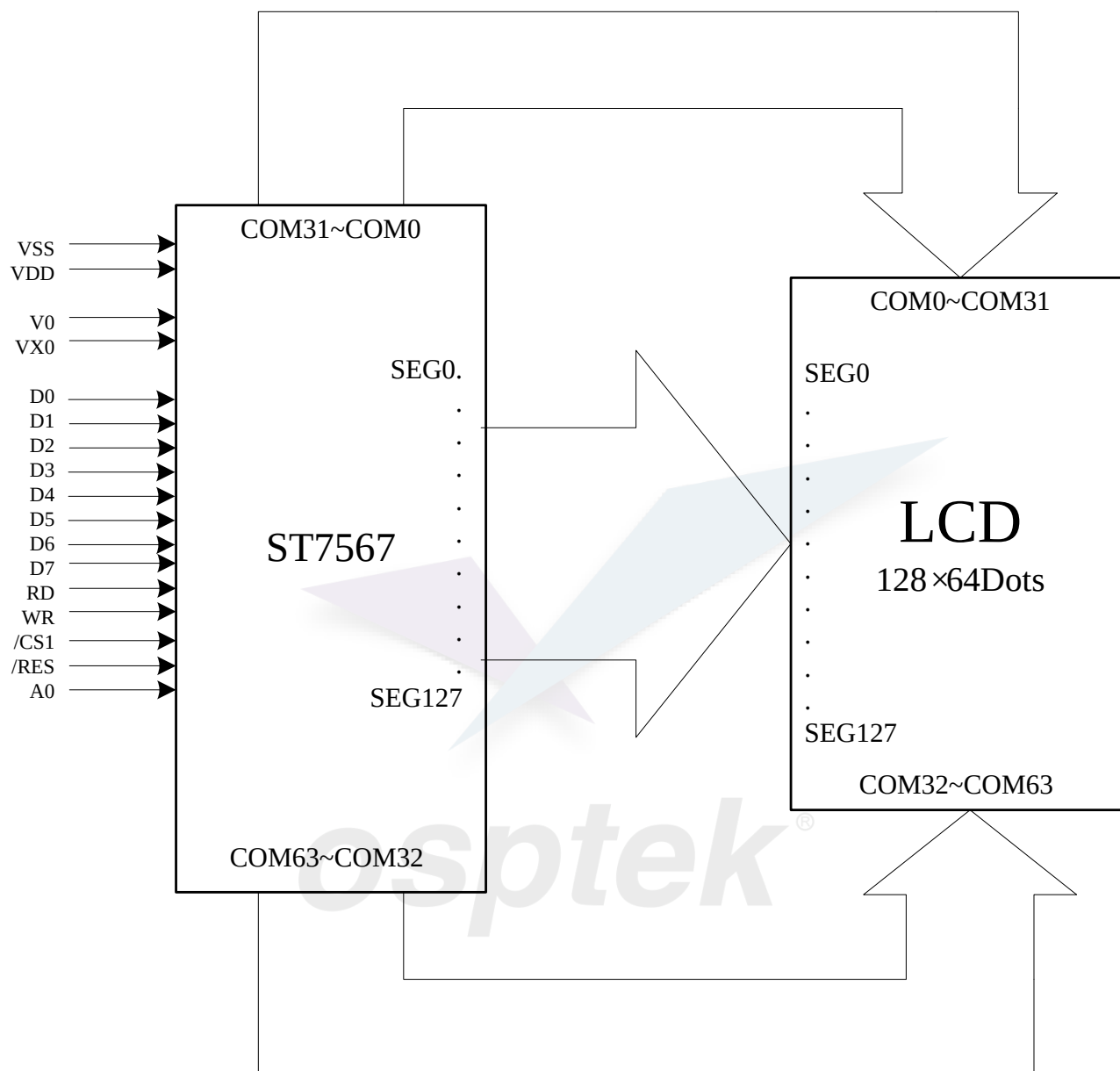
Shenzhen Osprey Optoelectronics Technology Co., Ltd.

<https://www.osptek.com>

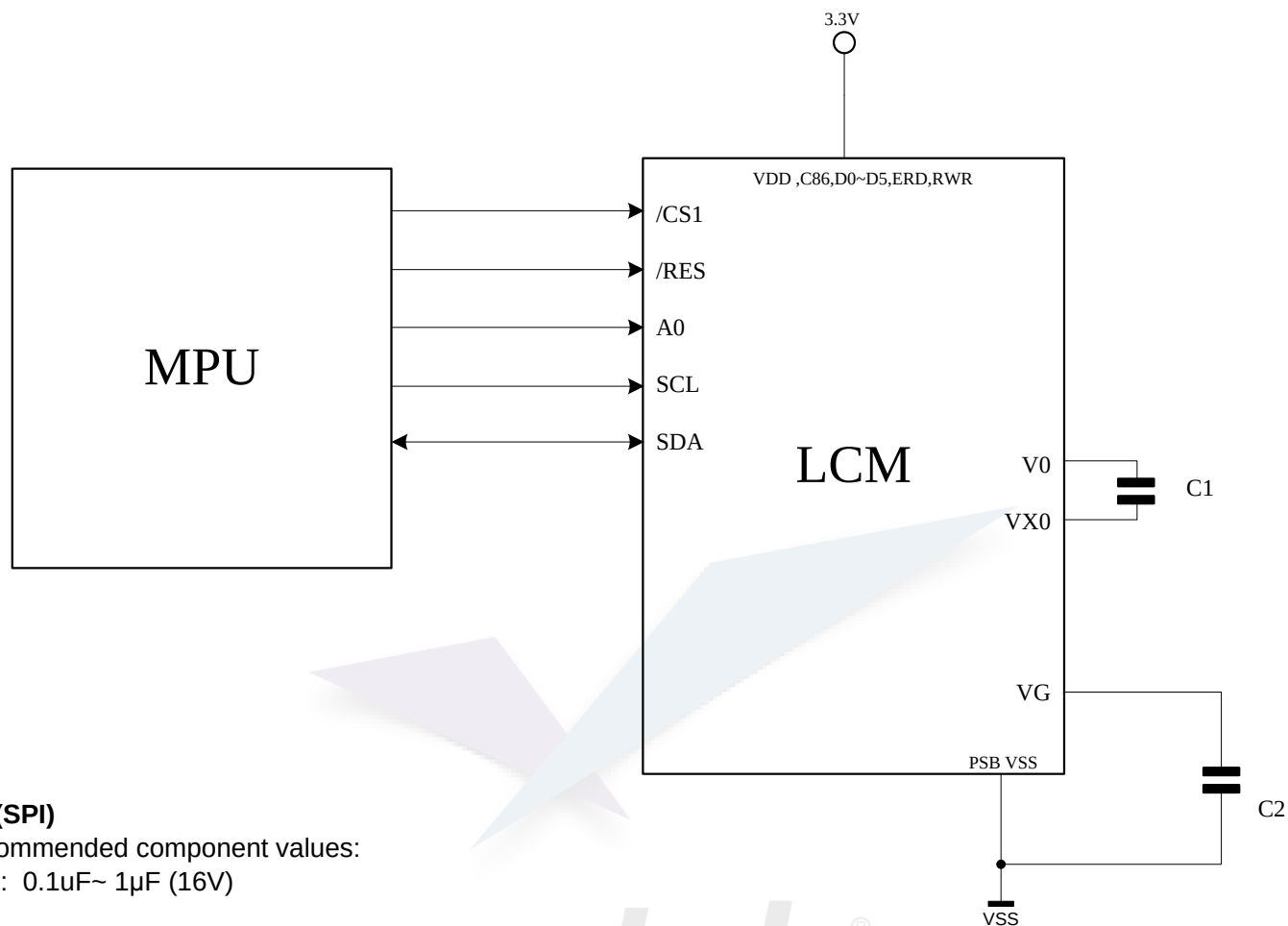
6. PIN CONNECTIONS

Pin No.	Pin Out	Description
1	CS1	Chip select input pin.
2	RESET	Reset input pin; When /RES is set to "L", Settings are initialized.
3	RS(A0)	Register select input pin; A0="H": display; A0="L": control data.
4	RW(W/R)	Read/Write execution control pin.
5	RD(E)	Read/Write execution control pin.
6	D0	8-bit parallel interface. When using serial interface: 4-LINE D7=SDA : Serial data input. D6=SCL : Serial clock input. D[5:0] are not used and should connect to "H" by VDD1 or VDDH. When CSB is non-active (CSB="H"), D[7:0] pins are high impedance.
7	D1	
8	D2	
9	D3	
10	D4	
11	D5	
12	D6	
13	D7	
14	VDD	Power supply for analog circuit.
15	VSS	Ground.
16	NC	No connection.
17	NC	No connection.
18	NC	No connection.
19	V0	V0 is the LCD driving voltage for common circuits at negative frame.
20	XV0	XV0 is the LCD driving voltage for common circuits at negative frame.
21	NC	No connection.
22	NC	No connection.
23	VG	VG is the LCD driving voltage for segment circuits.
24	NC	No connection.
25	NC	No connection.
26	NC	No connection.
27	NC	No connection.
28	P/S	PSB selects the interface type: Serial or Parallel. PSB="H", 8BIT PSB="L", 4 LINE (SPI)

7. BLOCK DIAGRAM



8. APPLICATION CIRCUIT EXAMPLE



Note (SPI)

- Recommended component values:
C1, C2: 0.1uF~ 1μF (16V)

电容建议用 1uF/25V, X7R 或 X5R

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9. ABSOLUTE MAXIMUN RATING

ELECTRICAL ABSOLUTE MAXIMUM RATINGS

ITEM	SYMBOL	MIN	MAX	UNIT
POWER SUPPLY FOR LOGIC	VDD-VSS	-0.3	3.6	V
POWER SUPPLY FOR LCD DRIVE	VLCD	-0.3	13.0	V
INPUT VOLTAGE	VIN	-0.3	VDD+0.3	V
POWER SUPPLY FOR LED	VA-VK	-0.3	4.0	V

ENVIRONMENTAL ABSOLUTE MAXIMUM RATINGS

ITEM	OPERATING		STORAGE		UNIT	COMMENT
	MIN	MAX	MIN	MAX		
AMBIENT TEMPERATURE	-20	+70	-30	+80	℃	
HUMIDITY	NOTE(1)		NOTE(1)		--	WITHOUT CONDENSTION
VIBRATION (M/S^2)	/	/	/	/	--	SEE "ITEMS OF RELIABILITY"
TEMPERATURE CYCLING TEST	/	/	/	/	--	SEE "ITEMS OF RELIABILITY"

10. ELECTRICAL CHARACTERISTICS (Vss=0V)

Item	Symbol	Condition	Min.	Typ	Max.	Unit	note
Power Supply for Logic	V _{DD} -V _{SS}	T _a =25℃	2.4	3.0	3.3	Volt	
	-	-	-	-	-	-	
Power Supply for LED	A-K	T _a =25℃	--	3.5	3.7	Volt	
Input Voltage	V _{IL}	V _{dd} =3V±5%	V _{SS}		0.2 V _{DD}	Volt	
	V _{IH}		0.8 V _{DD}	-	V _{DD}	Volt	
Output Voltage	V _{OL}	V _{dd} =3V±5%	V _{SS}	-	0.2 V _{DD}	Volt	
	V _{OH}		0.8 V _{DD}	-	V _{DD}	Volt	
LCD driveVoltage(recommended Voltage)	V ₀ -XV ₀	T _a =0℃	--	--	--	Volt	VDD 3.0V
		T _a =25℃	8.6	8.8	9.0		
		T _a =50℃	--	--	--		
Power Supply Current for LCM	I _{DD}	V _{DD} =3.0V	-	150	300	uA	-
	I _{LED}	V _{LED} = 3.0 V	--	14	40	mA	None1

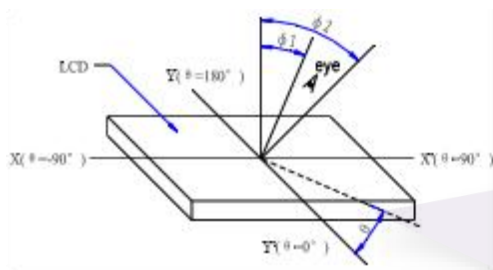
Note1:Backlight Electrical-Optical Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit	Conditon
Luminance	L _v	200	300	--	CD/m²	If=14 mA
Range	x	0.27	0.30	0.33	-	
	y	0.27	0.30	0.33	-	
Brightness uniformity	△%	70	-	-	%	min / max * 100%

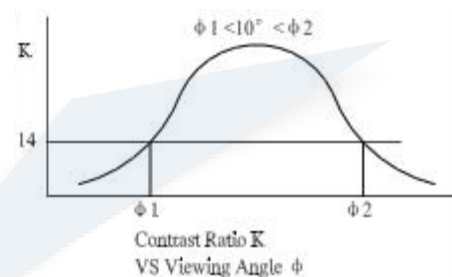
11. ELECTRO-OPTICAL CHARACTERISTICS

Item	Symbol	Condition	Min.	Typ	Max.	Unit	note
Viewing angle range	$\Theta 2 - \Theta 1$	$T_a = 25^\circ\text{C}$	20	-	-	Deg	$K=1.4$ A,B
	Φ		-	-	-		
Rise Time	T_r	$T_a = 25^\circ\text{C}$	-	200	300	ms	$\Phi=10$ $\Theta=0$ C
		$T_a = 0^\circ\text{C}$	-	-	-		
Fall Time	T_f	$T_a = 25^\circ\text{C}$	-	230	350		
		$T_a = 0^\circ\text{C}$	-	-	-		
Contrast	Cr	$T_a = 25^\circ\text{C}$	-	5	-	-	$\Phi=10$ $\Theta=0$ D

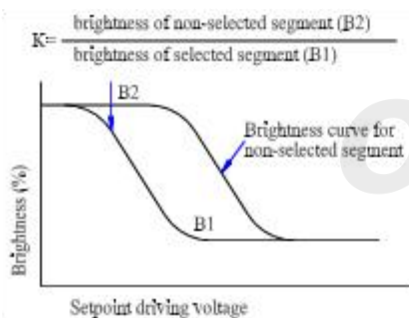
11.1 Definition of angle θ and ϕ



11.2 Definition of viewing angle $\phi 1$ and $\phi 2$

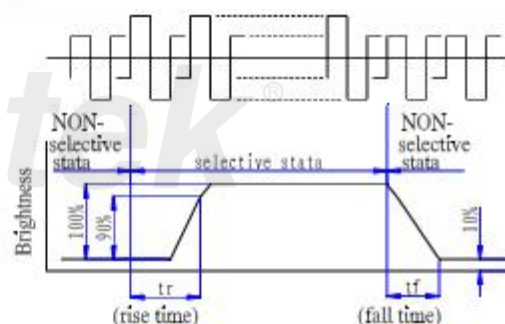


11.3 Definition of contrast "K"



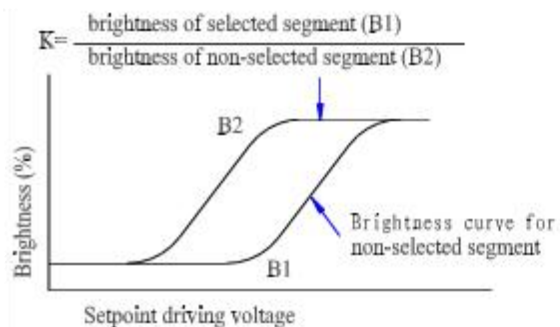
POSITIVE TYPE

11.4 Definition of optical response

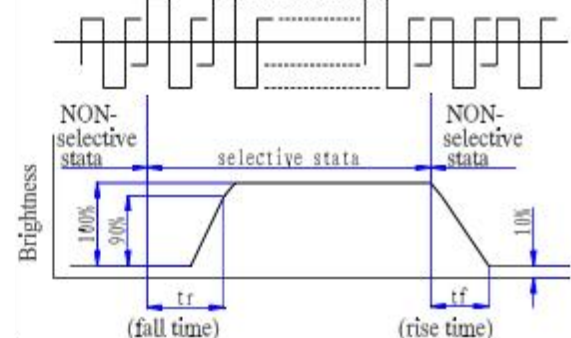


POSITIVE TYPE

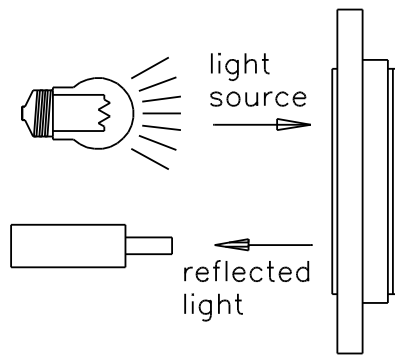
11.5 Definition of contrast "K"



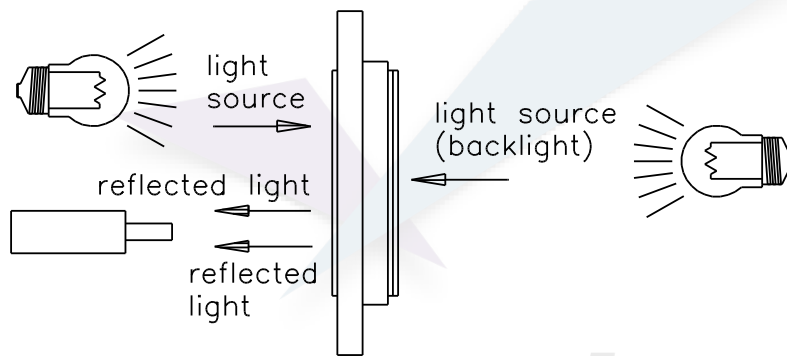
11.6 Definition of optical response



11.7 DESCRIPTION OF MEASURING EQUIPMENT



Reflective type



Transreflective type

12. CONTROLLER ELECTRICAL CHARACTERISTICS

DC CHARACTERISTICS

VSS=0V; Tamb = -30°C to +85°C; unless otherwise specified.

Item	Symbol	Condition		Rating			Unit	Applicable Pin
				Min.	Typ.	Max.		
Operating Voltage (1)	VDD1			1.7	—	3.3	V	VDD1
Operating Voltage (2)	VDD2			2.4	—	3.3	V	VDD2
Operating Voltage (3)	VDD3			2.4	—	3.3	V	VDD3
Input High-level Voltage	V _{IHC}			0.7 x VDD1	—	VDD1	V	MPU Interface
Input Low-level Voltage	V _{ILC}			VSS1	—	0.3 x VDD1	V	MPU Interface
Output High-level Voltage	V _{OHC}	I _{OUT} =1mA, VDD1=1.8V		0.8 x VDD1	—	VDD1	V	D[7:0]
Output Low-level Voltage	V _{OLC}	I _{OUT} =-1mA, VDD1=1.8V		VSS1	—	0.2 x VDD1	V	D[7:0]
Input Leakage Current	I _{LI}			-1.0	—	1.0	μA	MPU Interface
Output Leakage Current	I _{LO}			-3.0	—	3.0	μA	MPU Interface
Liquid Crystal Driver ON Resistance	R _{ON}	Ta=25°C	Vop=8.5V, ΔV=0.85V	—	0.6	0.8	KΩ	COMx
			VG=1.9V, ΔV=0.19V	—	1.3	1.5	KΩ	SEGx
Frame Frequency	FR	Duty=1/65, Vop=8.5V Ta = 25°C		70	75	80	Hz	

Current consumption: During Display, with internal power system, current consumed by whole IC (bare die).

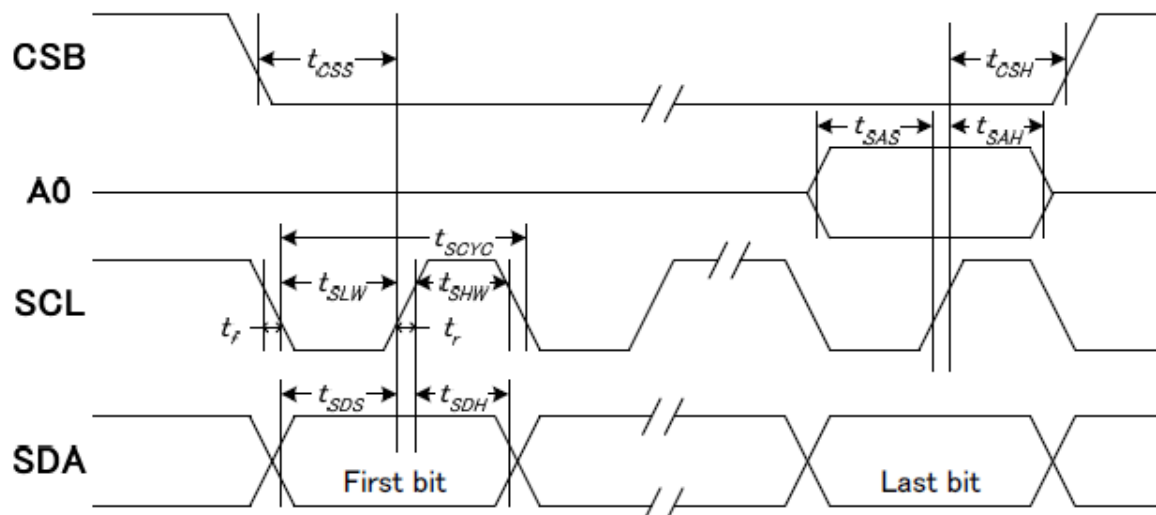
Test Pattern	Symbol	Condition	Rating			Unit	Note
			Min.	Typ.	Max.		
Display Pattern: SNOW (Static)	ISS	VDD1=VDD2=VDD3=3.0V, Booster X5 V _{OP} = 8.5 V, Bias=1/9 Ta=25°C	—	150	300	μA	
Display OFF	ISS	VDD1=VDD2=VDD3=3.0V, Booster X5 V _{OP} = 8.5 V, Bias=1/9 Ta=25°C	—	95	190	μA	
Power Down	ISS	VDD1=VDD2=VDD3=3.0V, Ta=25°C	—	8	16	μA	

Note:

- The Current Consumption is DC characteristics

13. TIMING CHARACTERISTICS

System Bus Timing for 4-Line Serial Interface



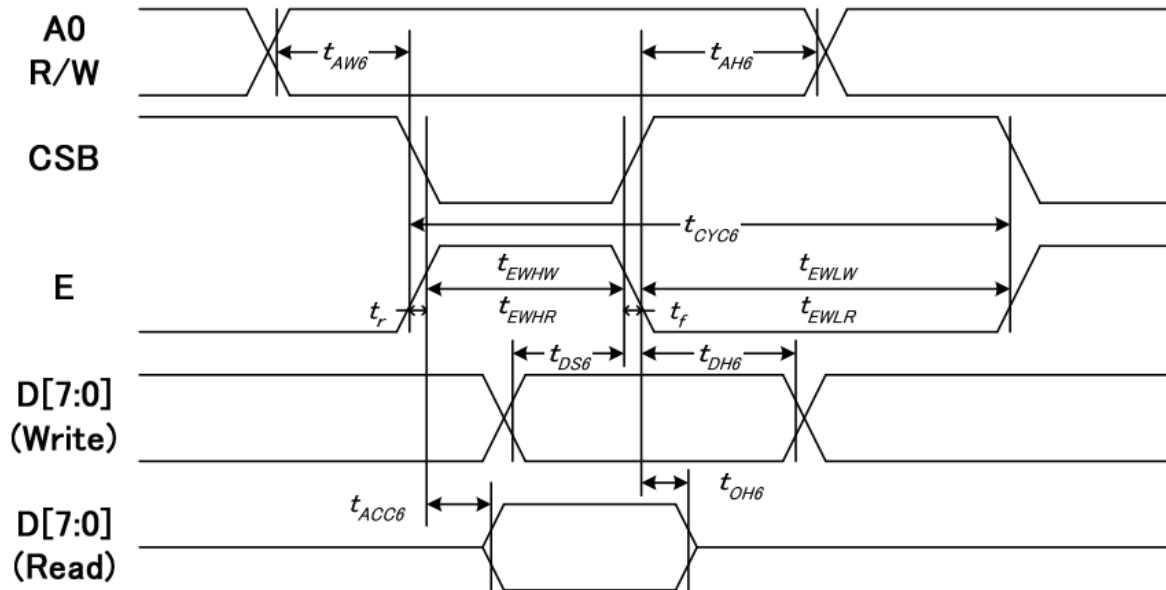
(VDD1 = 3.3V, Ta = 25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Serial clock period	SCLK	tSCYC		50	—	ns
SCLK "H" pulse width		tSHW		25	—	
SCLK "L" pulse width		tSLW		25	—	
Address setup time	A0	tSAS		20	—	
Address hold time		tSAH		10	—	
Data setup time	SDA	tSDS		20	—	
Data hold time		tSDH		10	—	
CSB-SCLK time	CSB	tCSS		20	—	
CSB-SCLK time		tCSH		40	—	

(VDD1 = 2.8V, Ta = 25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Serial clock period	SCLK	tSCYC		100	—	ns
SCLK "H" pulse width		tSHW		50	—	
SCLK "L" pulse width		tSLW		50	—	
Address setup time	A0	tSAS		30	—	
Address hold time		tSAH		20	—	
Data setup time	SDA	tSDS		30	—	
Data hold time		tSDH		20	—	
CSB-SCLK time	CSB	tCSS		30	—	
CSB-SCLK time		tCSH		60	—	

System Bus Timing for 6800 Series MPU



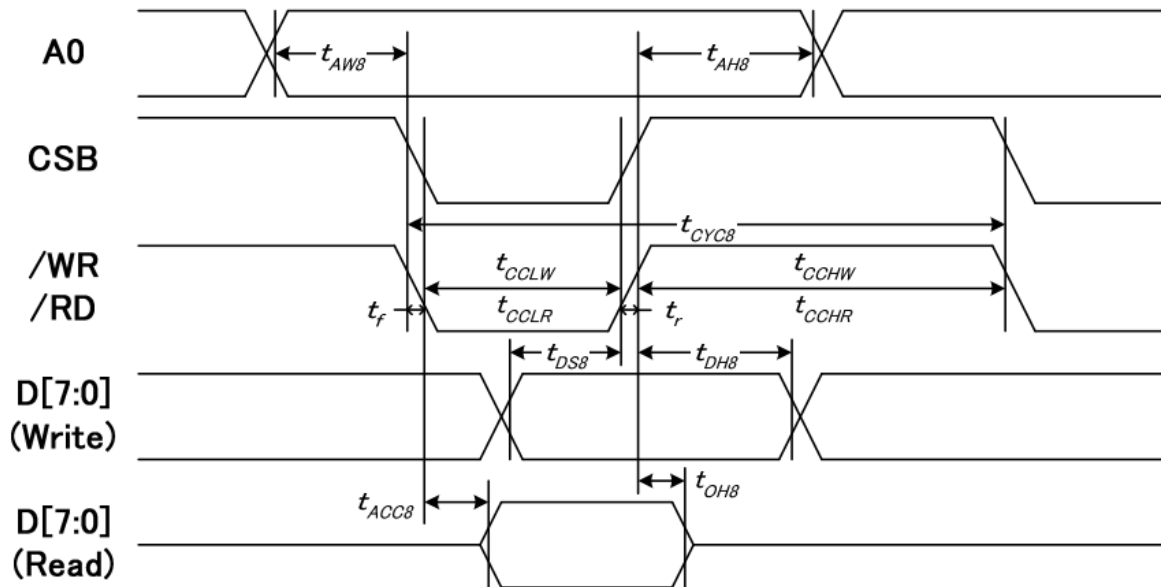
(VDD1 = 3.3V , Ta = 25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Address setup time	A0	tAW6		0	—	ns
Address hold time		tAH6		10	—	
System cycle time	E	tCYC6		240	—	
Enable L pulse width (WRITE)		tEWLW		80	—	
Enable H pulse width (WRITE)		tEHLW		80	—	
Enable L pulse width (READ)		tEWLR		80	—	
Enable H pulse width (READ)		tEHLR		140	—	
Write data setup time	D[7:0]	tDS6		40	—	
Write data hold time		tDH6		10	—	
Read data access time		tACC6	CL = 16 pF	—	70	
Read data output disable time		tOH6	CL = 16 pF	5	50	

(VDD1 = 2.8V , Ta = 25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Address setup time	A0	tAW6		0	—	ns
Address hold time		tAH6		0	—	
System cycle time	E	tCYC6		400	—	
Enable L pulse width (WRITE)		tEWLW		220	—	
Enable H pulse width (WRITE)		tEHLW		180	—	
Enable L pulse width (READ)		tEWLR		220	—	
Enable H pulse width (READ)		tEHLR		180	—	
Write data setup time	D[7:0]	tDS6		40	—	
Write data hold time		tDH6		20	—	
Read data access time		tACC6	CL = 16 pF	—	140	
Read data output disable time		tOH6	CL = 16 pF	10	100	

System Bus Timing for 8080 Series MPU



(VDD1 = 3.3V, Ta = 25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Address setup time	A0	t_{AW8}		0	—	ns
Address hold time		t_{AH8}		10	—	
System cycle time	/WR	t_{CYC8}		240	—	
/WR L pulse width (WRITE)		t_{CCLW}		80	—	
/WR H pulse width (WRITE)	RD	t_{CCHW}		80	—	
/RD L pulse width (READ)		t_{CCLR}		140	—	
/RD H pulse width (READ)	D[7:0]	t_{CCHR}		80	—	
WRITE Data setup time		t_{DS8}		40	—	
WRITE Data hold time		t_{DH8}		20	—	
READ access time		t_{ACC8}	CL = 16 pF	—	70	
READ Output disable time		t_{OH8}	CL = 16 pF	5	50	

(VDD1 = 2.8V, Ta = 25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Address setup time	A0	t_{AW8}		0	—	ns
Address hold time		t_{AH8}		0	—	
System cycle time	/WR	t_{CYC8}		400	—	
/WR L pulse width (WRITE)		t_{CCLW}		220	—	
/WR H pulse width (WRITE)	RD	t_{CCHW}		180	—	
/RD L pulse width (READ)		t_{CCLR}		220	—	
/RD H pulse width (READ)	D[7:0]	t_{CCHR}		180	—	
WRITE Data setup time		t_{DS8}		40	—	
WRITE Data hold time		t_{DH8}		20	—	
READ access time		t_{ACC8}	CL = 16 pF	—	140	
READ Output disable time		t_{OH8}	CL = 16 pF	10	100	

14. DISPLAY COMMANDS

The display commands shown below control the internal state of the LCD driver ICs. Commands are sent from CPU to LCD module for the display control.

INSTRUCTION	A0	R/W (RWR)	COMMAND BYTE								DESCRIPTION
			D7	D6	D5	D4	D3	D2	D1	D0	
(1) Display ON/OFF	0	0	1	0	1	0	1	1	1	D	D=1, display ON D=0, display OFF
(2) Set Start Line	0	0	0	1	S5	S4	S3	S2	S1	S0	Set display start line
(3) Set Page Address	0	0	1	0	1	1	Y3	Y2	Y1	Y0	Set page address
(4) Set Column Address	0	0	0	0	0	1	X7	X6	X5	X4	Set column address (MSB)
	0	0	0	0	0	0	X3	X2	X1	X0	Set column address (LSB)
(5) Read Status	0	1	0	MX	D	RST	0	0	0	0	Read IC Status
(6) Write Data	1	0	D7	D6	D5	D4	D3	D2	D1	D0	Write display data to RAM
(7) Read Data	1	1	D7	D6	D5	D4	D3	D2	D1	D0	Read display data from RAM
(8) SEG Direction	0	0	1	0	1	0	0	0	0	MX	Set scan direction of SEG MX=1, reverse direction MX=0, normal direction
(9) Inverse Display	0	0	1	0	1	0	0	1	1	INV	INV =1, inverse display INV =0, normal display
(10) All Pixel ON	0	0	1	0	1	0	0	1	0	AP	AP=1, set all pixel ON AP=0, normal display
(11) Bias Select	0	0	1	0	1	0	0	0	1	BS	Select bias setting 0=1/9; 1=1/7 (at 1/65 duty)
(12) Read-modify-Write	0	0	1	1	1	0	0	0	0	0	Column address increment: Read:+0, Write:+1
(13) END	0	0	1	1	1	0	1	1	1	0	Exit Read-modify-Write mode
(14) RESET	0	0	1	1	1	0	0	0	1	0	Software reset
(15) COM Direction	0	0	1	1	0	0	MY	-	-	-	Set output direction of COM MY=1, reverse direction MY=0, normal direction
(16) Power Control	0	0	0	0	1	0	1	VB	VR	VF	Control built-in power circuit ON/OFF
(17) Regulation Ratio	0	0	0	0	1	0	0	RR2	RR1	RR0	Select regulation resistor ratio
(18) Set EV	0	0	1	0	0	0	0	0	0	1	Double command!! Set electronic volume (EV) level
	0	0	0	0	EV5	EV4	EV3	EV2	EV1	EV0	
(19) Set Booster	0	0	1	1	1	1	1	0	0	0	Double command!! Set booster level: BL=0: 4X BL=1: 5X
	0	0	0	0	0	0	0	0	0	BL	
(20) Power Save	0	0	Compound Command								Display OFF + All Pixel ON
(21) NOP	0	0	1	1	1	0	0	0	1	1	No operation
(22) Test	0	0	1	1	1	1	1	1	1	-	Do NOT use. Reserved for testing.

Note: Symbol "-" means this bit can be "H" or "L".

15. QUALITY LEVEL

Inspection conditions

Environmental conditions

The environmental conditions for inspection shall be as follows: Room temperature: 22 ± 3 °C ; Humidity: $50 \pm 10\%$ RH

The external visual inspection

The inspection shall be performed by using a single 20W fluorescent lamp for illumination and the distance from LCD to eyes of the inspector should be 30cm or more.

Classification of defects

A major defect

A major defect refers to A defect which may substantially degrade usability for product applications.

Minor defect

A Minor defect refers to A defect which is not considered to substantially degrade product application or A defect which deviates from existing standards almost unrelated to the effective use of the product or its operation

Sampling procedures for each items acceptance level table

Defect type	Sampling procedures	AQL
Major defect	MIL-STD-105D Inspection level1 normal inspection Single sample inspection	1.0
Minor defect	MIL-STD-105D Inspection level1 normal inspection Single sample inspection	2.5

Life time

50,000Hrs(25°C in the room without ray of sun)

Items of reliability

ITEM	CONDITIONS	CRITERION
High temperature operation test	$+70^{\circ}\text{C}$ \ 120 hours	1. It judged at room temperature after 1 hours to be good as appearance and electrical test is normal after the experiment. 2. Current consumption should within the specification of Approval sheet Electro-optical characteristics
Low temperature operation test	-20°C \ 120 hours	5-10pcs
High temperature/humidity storage test	$+80^{\circ}\text{C}$, $80\% \pm 10\%$ RH \ 120 hours	
High temperature storage test	$+80^{\circ}\text{C}$ \ 120 hours	
Low temperature storage test	-30°C \ 120 hours	

Temperature cycling test	-20°C (30 min) $\downarrow$ $\uparrow$ 25°C (5 min) $\downarrow$ $\uparrow$ 70°C (30 min)	
	CYCLES: 10	
Vibration	Random Wave: 10 ~ 50 Hz Each Direction (x, y, z): 30 Min.	

Cosmetic criteria of LCD screen

Cosmetic criteria of LCD screen

DEFECT	JUDGEMENT CRITERION		
Spots	Size d (mm)		Acceptable quantity in active area
	d≤0.1		Disregard
	0.1<d≤0.2		6
	0.2<d≤0.3		2
	d>0.3		0
	Note: d = (Length + Width)/2		
Polarizer Bubbles	Size d (mm)		Acceptable quantity in active area
	d≤0.3		Disregard
	0.3<d≤1.0		3
	1.0<d≤1.5		1
	d>1.5		0
	Note: d = (Length + Width)/2		
Lines	Width W (mm) Length L (mm)		Acceptable quantity in active area
	W ≤ 0.02		Disregard
	0.02< W≤0.05	L ≤ 5.0	6
		L > 5.0	0
	0.05<W≤0.1	L ≤ 2.0	6
		L > 2.0	0
	W > 0.1		See criteria for spots
Testing conditions: 20W fluorescent lamp at 30 cm distance at normal viewing angle			

16. PRECAUTIONS

Static charge

Since this LCD module contains CMOS LSI that are sensitive to static charge, care must be taken when handling it.

Power on sequence

1. Input signals should not be applied to the LCD module before the logic system voltage has reached the specified voltage. If the above sequence is not kept, the LCD module might be permanently damaged.
2. When connecting the power supply, connect the LCD bias voltage after connecting the logic system voltage.
3. When disconnecting the power supply, disconnect the logic system voltage after the LCD bias voltage.
4. It is recommended to connect a serial resistor or fuse to the LCD bias power supply of the system as a current limiter. The value of the resistor depends on the kind of LCD used, but is typically 50~100 Ω

Operation

1. It is essential to drive the LCD within the specified voltage limits, since a higher driving voltage than allowed causes a shorter LCD lifetime. Under these circumstances, electrochemical reactions will result in undesirable deterioration of the LCD.
2. The response time of the LC fluid is considerably longer at low temperature than in the normal operating temperature range. On the other hand, the LCD will show a dark blue color at high temperatures. Those phenomena do not indicate a malfunction or defect of the LCD. Back at normal temperatures, the LCD will return to its original behavior.
3. If the display area is pressed hard during operation, some abnormal display patterns might appear. However, the display will resume normal operation after turning the module off and on.
4. Moisture on the terminals could cause an electrochemical reaction resulting in an open terminal connection. If the environmental temperature is higher than 50°C, it is required that the relative humidity is 50% or less.

Long-time storage

For long-term storage the following methods are highly recommended:

1. Store the product in a polyethylene bag with a sealed opening to prevent fresh air entering from the outside. Placing it with a desiccant is not necessary.
2. Store the product in a dark place, with the temperature in the range from -10°C to 50°C.
3. Keep the sensitive polarizer surface of the LCD panels clear of any contact. We recommend using the container that was used by OSP to deliver the products.

Cleaning of product

To clean the product make sure to use absorbent cotton cloth or other soft material like chamois. Make sure to rub it gently and do not use chemicals when cleaning.

17. PACKAGE INFORMATION

No.	Item	Packaging	Material	Quantity
		Model	Dimensions (mm)	
1	LCM	-	-	-
2	POF	-	-	-
3	TRAY	-	-	-
4	SMALL BOX	-	385.0×315.0×200.0	2
5	BIG BOX	-	398.0×331.0×430.0	1

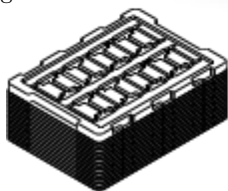
Use empty tray



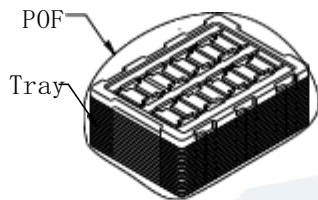
Put products into the tray



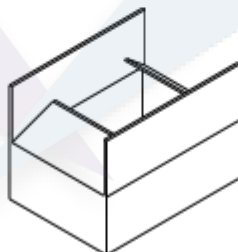
Tray stacking



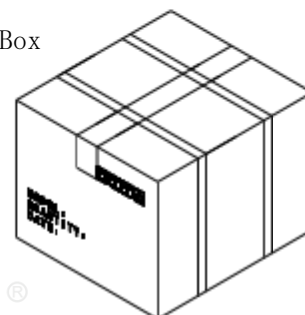
POF
Tray



Small Box



Big Box



osptek®